

The action potential

The written notes for the action potential walkthrough. Week 4 competencies 13 to 16.

To print: these notes as a two-column PDF.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the slides.

The key numbers

Point	Voltage	What sets it
Rest	-70 mV	Mostly potassium leak (resting potential walkthrough)
Threshold	about -55 mV	Sodium current in first outweighs potassium current out
Peak	about +30 mV	Sodium inactivation gates close and potassium channels open
After-hyperpolarization	about -80 mV	Potassium channels slow to close
Sodium's balance point	+60 mV	The ceiling the rise heads toward
Potassium's balance point	-90 mV	The floor the dip heads toward

The whole spike, from threshold back to rest, lasts a few milliseconds. The rise and the fall together take about 1 to 2 ms.

The two voltage-gated channels

Channel	Gates	At rest	When depolarized
Voltage-gated Na⁺ channel	Activation gate (outside), inactivation gate (inside)	Activation closed, inactivation open	Activation opens fast; inactivation closes a little later and stays closed until the membrane repolarizes
Voltage-gated K⁺ channel	One gate	Closed	Opens slowly, at about the time sodium channels inactivate, and closes slowly

- Both channels are opened by depolarization. The difference that shapes the action potential is timing: sodium is fast, potassium is slow.
- A stimulus can open an activation gate. Nothing can force an inactivation gate open. It reopens only after the membrane repolarizes.

Reaching threshold

- A graded potential depolarizes the trigger zone and opens some voltage-gated sodium channels.

- Below threshold (example: a 10 mV push to -60 mV) the sodium coming in is less than the potassium leaking out. The membrane falls back to rest. Nothing fires.
- At threshold (a 15 mV push to about -55 mV) the sodium current in matches, then outweighs, the potassium current out.
- Threshold is a balance of two currents, not a fixed number.

1. Sodium comes in and the membrane depolarizes.
2. Depolarization opens more voltage-gated sodium channels.
3. More sodium comes in, and the membrane depolarizes further.
4. Back to step 2. This self-feeding loop is positive feedback, and once it starts the stimulus no longer matters.

The phases, in order

Phase	Voltage	Na ⁺ channels	K ⁺ channels	Ion movement
Rest	-70 mV	Closed (activation gate closed)	Closed	Leak only
Rising phase (depolarization)	-55 to about +30 mV	Open	Starting to open	Na ⁺ rushes in
Peak (overshoot above 0 mV)	about +30 mV	Inactivation gates close	Open	Na ⁺ stops; K ⁺ starts to leave
Falling phase (repolarization)	+30 back to -70 mV	Inactivated	Open	K ⁺ flows out
After-hyperpolarization	down to about -80 mV, then back to rest	Resetting: activation closes, inactivation opens	Still open, closing slowly	Extra K ⁺ out moves the membrane toward -90 mV

- The rise heads toward sodium's balance point (+60 mV) because the membrane is now most permeable to sodium, the same rule as the resting potential.
- It stops near +30 mV because the sodium inactivation gates close and the potassium channels open.
- Potassium leaving brings the voltage down. The sodium-potassium pump does not.
- The dip below rest happens because the potassium channels are slow to close, so the membrane is briefly more permeable to potassium than at rest.

Threshold

The voltage at which sodium current in first outweighs potassium current out and the action potential takes off; about -55 mV here.

Positive feedback

A loop in which the result of a step makes the step happen more: depolarization opens sodium channels, which cause more depolarization.

Rising phase (depolarization)

The fast climb from threshold to the peak, carried by sodium coming in.

Overshoot

The part of the action potential above 0 mV, when the inside is positive.

Falling phase (repolarization)

The fall from the peak back toward rest, carried by potassium going out.

After-hyperpolarization

The brief dip below rest after the spike, caused by potassium channels that are slow to close.

How many ions move

- Only a tiny fraction of the ions cross in one action potential. Inside the cell, sodium stays about 15 mM and potassium about 150 mM.
- The gradients are still there after a spike, so a neuron can fire again right away without waiting for the pump.
- The pump keeps concentrations steady over the long run, across thousands of action potentials. It does not repolarize the membrane after each spike.

All or none

- Below threshold: no action potential at all. At or above threshold: a full-sized one.
- Example: pushes of 5 and 10 mV stay below threshold. Pushes of 16 and 30 mV both give action potentials that peak at about +30 mV.
- The size is set by the channels and sodium's balance point, not by the stimulus. A bigger push only reaches threshold a little sooner.

All or none

An action potential either happens at full size or does not happen; its size does not depend on the size of the stimulus.

Coding stimulus strength

- Because every spike is the same size, size cannot carry information.
- Frequency coding: a stronger stimulus makes a single neuron fire more often. A bigger graded potential reaches threshold again sooner after each spike. Example in the walkthrough: 2, 5 and 9 action potentials in the same time for weak, medium and strong.
- Population coding: a stronger stimulus activates more receptors, so more neurons fire. Example: 1, 2 and 4 neurons.
- A single axon can only use frequency coding.

Frequency coding

Signaling stimulus strength by how often a neuron fires.

Population coding

Signaling stimulus strength by how many neurons fire.

Refractory periods

Period	When	Channel state	Can a second action potential start?
Absolute refractory period	From the start of the spike until the sodium channels begin to reset, through most of the fall	Na ⁺ inactivation gates closed	No, no matter how strong the stimulus
Relative refractory period	During the dip below rest	Some Na ⁺ channels reset; extra K ⁺ channels still open	Yes, but it takes a stronger stimulus than usual, and the second spike is often a little smaller

- Example: during the dip the membrane is near -78 mV. The same 16 mV push that fired the first spike reaches only -62 mV, short of threshold.
- The absolute refractory period lasts about 1 to 2 ms, which sets a ceiling on firing rate: $1,000 \text{ ms} \div 2 \text{ ms} = 500$ per second, or $1,000 \text{ ms} \div 1 \text{ ms} = 1,000$ per second. Most neurons fire well below this.
- The refractory period also keeps the action potential traveling in one direction along the axon (conduction walkthrough).

Absolute refractory period

The time during which no stimulus of any size can start a second action potential, because the sodium inactivation gates are closed.

Relative refractory period

The time right after, during which a second action potential can start but only with a stronger stimulus than usual.

Patient: a drug that blocks potassium channels

- Dalfampridine blocks voltage-gated potassium channels. It is used to help people with multiple sclerosis walk.
- In multiple sclerosis, axons lose myelin, which exposes potassium channels that are normally covered.
- With the potassium channels blocked, the rise and peak barely change, the fall is much slower, and the dip below rest mostly disappears. The action potential gets wider.
- A wider action potential delivers more charge to the next stretch of axon, which helps the signal cross a stretch that has lost its myelin.
- Blocking these channels throughout the brain can make neurons fire too easily, so the dose is kept low: too much can cause seizures.